

LECTURE 8 (SEPTEMBER 30TH)

Theorem. Let $\{f_n\}$ be a sequence of 1-1 analytic functions on a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D . Then, either f is constant or f is 1-1 on D .

Proof. Let f be non-constant. Suppose that there exists $z_1, z_2 \in D$ such that $f(z_1) = f(z_2) = a$ and that the function is not 1-1. Then, there exists $r > 0$ such that:

- (i) $\bar{D}(z_1, r) \subset D$ and $\bar{D}(z_2, r) \subset D$
- (ii) By the identity theorem, $f(z) - a$ has no zeros in both $0 < |z - z_1| \leq r$ and $0 < |z - z_2| \leq r$. Denote simple circles $C_1 : |z - z_1| = r$ and $C_2 : |z - z_2| = r$. Let

$$m = \min_{z \in C_1 \cup C_2} |f(z) - a| > 0$$

where $C_1 \cup C_2$ is compact.

- (iii) $\bar{D}(z_1, r) \cap \bar{D}(z_2, r) = \emptyset$

By the uniform convergence of $\{f_n\}$ on $C_1 \cup C_2$, there is N such that

$$|f_N(z) - f(z)| < m$$

for $z \in C_1 \cup C_2$. On C_1 , $|f_N(z) - f(z)| < m \leq |f(z) - a|$. By Rouché's theorem, $f(z) - a$ and $f_N(z) - a$ have the same numbers of zero inside C_1 . Also, $f(z) - a$ and $f_N(z) - a$ have the same number of zeros inside C_2 . Thus, $f_N(z) - a$ has zeros inside C_1 and inside C_2 . Thus, f_N is not 1-1. $\square$

Example. There are multiple cases where the first option holds. For example, take $f_n(z) = \frac{z}{n}$ and $f = 0$.

Theorem. (Riemann mapping theorem) If $D (\neq \mathbf{C})$ is a simply connected domain, then there is a 1-1 analytic function h from D onto U .

Proof. Let $\Sigma = \{f : D \rightarrow U \mid f \text{ is injective and analytic}\}$.

(I) We prove that $\Sigma \neq \emptyset$. To prove this, we use the prior result that if f is analytic on a simply connected domain D , such that $f(z) \neq 0$ for any $z \in D$, then there is g analytic in D such that $g^2 = f$ on D . Suppose that $w_0 \notin D$. Then there is g analytic on D such that $g^2(z) = z - w_0$. As $w^2 = z$ implies that $w = \sqrt{r}e^{i\theta/2}$, $g(D)$ does not intersect some half of the plane! Furthermore,

- (i) $0 \notin g(D)$, which is trivial.

- (ii) g is 1-1 on D , as if $g(z_1) = g(z_2)$, then $g^2(z_1) = g^2(z_2)$ and $z_1 - w_0 = z_2 - w_0$.
- (iii) If $w \in g(D)$, then $-w \notin g(D)$, as if $g(z_1) = -g(z_2)$ for distinct points, then $g^2(z_1) = g(z_2)^2$ and $z_1 = z_2$ which is a contradiction.
- (iv) If $a \in g(D)$, then there exists an $r > 0$ such that $D(a, r) \subset g(D)$ due to the open mapping theorem. In particular, $D(-a, r) \cap g(D) = \emptyset$ as if $w \in D(-a, r)$, then $|w + a| < r$ and $|-w - a| = |w + a| < r$ which implies that $-w \in D(a, r) \subset g(D)$.

Now, define $h : D \rightarrow \mathbb{C}$ by

$$h(z) = \frac{r}{g(z) + a}$$

Notice that $|g(z) + a| > r$ for all $z \in D$ and that $|h(z)| < 1$ for all $z \in D$. We see that if $h(z_1) = h(z_2)$.

$$\frac{r}{g(z_1) + a} = \frac{r}{g(z_2) + a}$$

and that $g(z_1) = g(z_2)$ with $z_1 = z_2$. We see that $h(z)$ is a 1-1 function from an arbitrary domain D to U !

(II) Let $\psi \in \Sigma$ such that $\psi(D) \neq U$. We claim that for every $z_0 \in D$, there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\psi'(z_0)|$. Suppose that $\alpha \in U \setminus \psi(D)$. Then, $\phi_\alpha \circ \psi$ is analytic on D with no zeros. Since D is simply connected, there is g analytic in D such that $g^2 = \phi_\alpha \circ \psi$. Let $g(z_0) = \beta$. Define $\psi_0 = \phi_\beta \circ g$. Then, $\psi_0 \in \Sigma$ and due to the Schwartz lemma, the claim holds. $\square$

LECTURE 9 (OCTOBER 2ND)

Proof. (I) Take any $w_0 \notin D$, and then there is g that is analytic in D such that $g^2(z) = z - w_0$. If $D(a, r) \subset g(D)$, then $D(-a, r) \cap g(D) = \emptyset$. Now define

$$h(z) = \frac{r}{g(z) + a}$$

then $h \in \Sigma$.

(II) We claim that if $\psi \in \Sigma$ and $\psi(D) \neq U$, then for every $z_0 \in D$, there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\psi'(z_0)|$. Let $\alpha \notin \psi(D)$, then $\phi_\alpha \circ \psi$ is analytic in D with no zeros in D . Thus, there is g analytic in D such that $g^2 = \phi_\alpha \circ \psi$ in D . Given $z_0 \in D$, if $\beta = g(z_0)$, then $\psi_0 = \phi_\beta \circ g$ satisfies the claim.

Notice how $g \in \Sigma$ since g is 1-1 in D (if $g(z_1) = g(z_2)$, then $(\phi_\alpha \circ \psi)(z_1) = (\phi_\alpha \circ \psi)(z_2)$ concluding that $z_1 = z_2$). Additionally denote the square function by $s(z) = z^2$. Then,

$$\phi_\alpha \circ \psi = g^2 = s \circ g$$

so that $\psi = \phi_{-\alpha} \circ s \circ g$. Additionally, $\psi_0 = \phi_{\beta} \circ g$ so that $g = \phi_{-\beta} \circ \psi_0$. Therefore,

$$\psi = \phi_{-\alpha} \circ s \circ \phi_{-\beta} \circ \psi_0$$

on D . Now we will show that the claimed inequality holds. Defining $F = \phi_{\alpha} \circ s \circ \phi_{-\beta}$ or $F : U \rightarrow U$ is analytic. By the Schwartz lemma,

$$|F'(0)| \leq 1$$

since F is not 1-1. From $\psi = F \circ \psi_0$, we have

$$\psi'(z_0) = F'(\psi_0(z_0))\psi_0'(z_0) = F'(0)\psi_0'(z_0)$$

Since ψ is 1-1, the derivative $\psi' \neq 0$ in D and we have proved the claim.

(III) We now claim that if we take any $z_0 \in D$ and let $\eta_0 = \sup\{|\psi'(z_0)| \mid \psi \in \Sigma\}$, there is $h \in \Sigma$ such that $|h'(z_0)| = \eta_0$. From this, as $h(D)$ doesn't conform to the condition of step (II), we will have that $h(D) = U$. The key method here is using "normal families". $\square$

Remark. On $\mathbf{R}^n$,

- (i) If G is a bounded infinite, then G has a limit point
- (ii) If $\{a_n\}$ is a bounded sequence, then $\{a_n\}$ has a convergent subsequence
- (iii) If K is closed and bounded, then every open cover of K has a finite subcover

However, consider the following. The Hilbert space $l^2(\mathbf{N})$ can be seen as the space closest to $\mathbf{R}^\infty$.

$$\mathbf{x} = (x_1, x_2, \dots) \in l^2(\mathbf{N}) \quad \text{if} \quad \|\mathbf{x}\|^2 = \sum_{n=1}^{\infty} x_n^2 < \infty$$

Defined this manner, $l^2(\mathbf{N})$ is a complete normed vector space. Consider $E = \{\mathbf{e}_n \mid n \in \mathbf{N}\} \subset l^2(\mathbf{N})$ where $\mathbf{e}_k = (0, \dots, 0, 1, 0, \dots)$ where only the k -th element is nonzero. This basis satisfies $\|\mathbf{e}_k\| = 1$.

- (i) Then, E is a bounded infinite subset of $l^2(\mathbf{N})$ which has no limit point.
- (ii) Also, E is a bounded sequence in $l^2(\mathbf{N})$ with no convergent subsequence.
- (iii) Lastly, E is closed and bounded in $l^2(\mathbf{N})$ but is not compact.

Definition. (Three types of convergences & boundednesses) If D is open and $f_n : D \rightarrow \mathbf{C}$, then we can define three types of convergences.

- (i) Pointwise convergence in D
- (ii) Uniform convergence in D

(iii) Uniform convergence in every compact subset of D

There are also three types of boundednesses of $\{f_n\}$ on D

- (i) Uniformly bounded on D , meaning that $|f_n| \leq M$ for all $n \in \mathbf{N}$ and $z \in D$.
- (ii) Pointwise bounded on D , meaning there is $M_z > 0$ such that $|f_n(z)| \leq M_z$ for every $n \in \mathbf{N}$.
- (iii) Uniformly bounded on every compact subset of D , meaning that for every compact subset K of D , there is $M_K > 0$ such that $|f(z)| \leq M_K$ for all $n \in \mathbf{N}$ and $z \in K$.

Theorem. (Arzela-Ascoli theorem) A separable metric space is space that has a countable dense subset and a metric space is a space endowed with a metric. If $\{f_n\}$ is a pointwise bounded and equicontinuous sequence of functions on a separable metric space, $\{f_n\}$ has a subsequence that converges uniformly on every compact subset.